

SciTables : A Dataset and Evaluation Framework for Complex Table-to-Text Generation

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Abstract

Generating coherent and factually grounded text from structured data is a core challenge in natural language generation, with applications in scientific communication, medical documentation, and automated reporting. Existing datasets primarily focus on open-domain or simplified table formats, limiting progress in more complex, high-stakes domains. We present a new dataset and framework **SciTables** for scientific table-to-text generation, addressing the gap in existing resources that focus largely on open-domain or simplified table formats. Our dataset is constructed from peer-reviewed Computer Science papers on arXiv (2017–2023) and features complex tables rich in numeric and symbolic content paired with naturally occurring textual descriptions. Unlike prior work relying on manual annotation, we develop a scalable, semi-automated pipeline to extract, clean, and align tables with their associated text, preserving domain-specific language and minimizing annotation cost. The resulting benchmark poses realistic challenges for current models, enabling rigorous evaluation of factuality, coherence, and domain adaptation. We conduct extensive experiments with state-of-the-art generation models and assessment strategies—including LLMs-as-judges and embedding-based semantic metrics—offering insights into model performance in high-precision scientific settings. Our work provides a foundation for developing more faithful and informative table-to-text systems in specialized domains.

1 Introduction

Generating coherent and factually grounded text from structured inputs is a core challenge in natural language generation (NLG) and a critical capability for many applications, including automated report generation, business intelligence, scientific publishing platforms, clinical documentation systems etc. In recent years, there has been notable progress in

data-to-text generation, much of this progress has been driven by the availability of large-scale, high-quality datasets in domains such as sports (Chen and Mooney, 2008; Wiseman et al., 2017), restaurants (Novikova et al., 2017), weather (Liang et al., 2009), biographies (Banik et al., 2013; Lebre et al., 2016), and open-domain settings (Parikh et al., 2020). However, in high-stakes domains like healthcare, pharmaceuticals, and scientific R&D, the challenge remains: structured data such as experimental tables, clinical trial results, or lab measurements must be translated into accurate and coherent natural language summaries. Yet, there is a striking lack of high-quality, large-scale datasets to support this task.

This gap limits the deployment of data-to-text systems in real-world setting, where outputs must not only be fluent, but also factually grounded, interpretable, and domain-specific. Creating such datasets at scale is expensive and often infeasible, requiring expert annotation to ensure that descriptions are faithful to underlying data. Manual annotation is costly, time-consuming, and often inconsistent, especially in domains that require specialized knowledge. These limitations can result in outputs that are not faithful to the source data—containing hallucinations, imprecise statements, or irrelevant information. These issues are especially problematic in high-stakes applications, where decisions may depend on the generated summaries.

To address these challenges, we introduce a new dataset and framework **SciTables** for scientific table-to-text generation. Our approach leverages a scalable, semi-automated pipeline that extracts and aligns complex tables from peer-reviewed research articles with high-quality textual descriptions. These scientific tables often contain rich, multi-dimensional data—posing significant challenges for current generation models and offering a rigorous testbed for benchmarking. By reducing annotation cost while maintaining factual consistency,

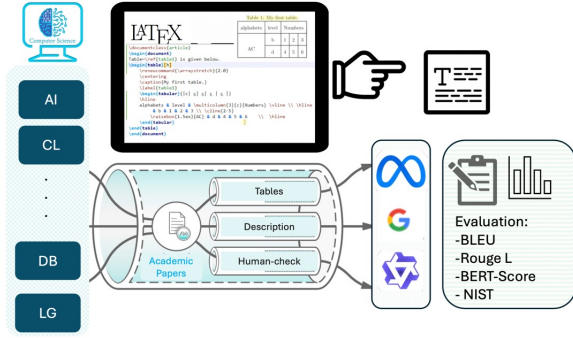


Figure 1: Flow of work

our framework enables practical dataset construction at scale. In addition to dataset creation, we conduct a comprehensive evaluation of state-of-the-art generation models, using both LLMs-as-judges and embedding-based semantic similarity metrics to assess the quality and faithfulness of generated outputs.

Our contributions are threefold: (1) we introduce a domain-specific dataset for complex table-to-text generation grounded in scientific and technical content; (2) we propose a scalable, pipeline for dataset creation with quality control; and (3) we benchmark models on this dataset, offering insights into the challenges of deploying faithful generation systems in real-world settings.

2 Related Work

Domain Specific Datasets: Table-to-text generation has seen considerable progress with the release of large-scale datasets in open and semi-structured domains. Notably, ToTTo (Parikh et al., 2020) and LogicNLG (Chen et al., 2020) have enabled model development at scale. These datasets, primarily based on Wikipedia, cover diverse topics and provide structured tables aligned with natural language descriptions. However, the tables tend to be relatively simple, with limited numeric or symbolic content, and are less suited to high-stakes domains requiring complex reasoning. Domain-specific datasets have also been introduced in narrower contexts. For instance, early work explored weather forecasts (Liang et al., 2009), restaurant recommendations (Novikova et al., 2017), sports game summaries (Chen and Mooney, 2008; Wiseman et al., 2017), entity descriptions (Qader et al., 2018; Wang et al., 2018) and biographies (Banik et al., 2013; Lebet et al., 2016). These datasets typically involve controlled language and limited variability in table structures.

Despite these advances, scientific and technical domains remain underrepresented. Scientific tables often involve dense numerical, symbolic, and domain-specific information, and their interpretation requires more than surface-level content alignment. This remains a notable gap in the literature—one that our work aims to address.

Scientific Table-to-Text Generation. Extending beyond existing domains, our work introduces a new dataset for table-to-text generation derived from scientific articles in the Computer Science section of arXiv. We collect tables and their corresponding natural language descriptions from a diverse set of papers published between 2017 and 2023. Figure 2 provides an overview of the subdomain distribution within our dataset.

Unlike prior datasets, our tables often contain dense numeric and symbolic data, accompanied by highly technical descriptions. This results in a more realistic and challenging benchmark for testing the capabilities of data-to-text models, especially with respect to logical reasoning, factual consistency, and domain adaptation. As such, our dataset serves as a valuable resource for evaluating the performance of large language models on structured scientific content.

Annotation Strategies and Automation. Constructing large-scale, high-quality datasets through manual annotation is expensive, time-consuming, and difficult to scale—particularly in specialized domains where expert knowledge is required. To mitigate this, our approach shifts toward automation. In contrast to the ToTTo dataset (Parikh et al., 2020), where annotators were instructed to rewrite contextually relevant sentences, we directly extract natural descriptive texts from scientific articles without manual rewriting. To ensure text quality, we apply a cleaning step that removes LaTeX notations, formatting commands, and other artifacts that can distort the semantic content or interfere with embedding-based processing. This process preserves the domain-specific terminology and natural linguistic structure, resulting in a more authentic and semantically aligned pairing between structured data and textual description.

Prior work relying on manual annotations often suffers from limited domain expertise, which may lead to reduced factual accuracy and alignment. Furthermore, the repeated use of similar phrasing or sentence structures by annotators can introduce stylistic bias and reduce linguistic diversity within the dataset. Our automated pipeline

Dataset	Pairs	Domain	Target Source	Annotation
WIKIBIO (Lebret et al., 2016)	400K	Biographies	Wikipedia	Automated
TOTTo (Parikh et al., 2020)	136K	Wikipedia (open-domain)	Wikipedia (Annotator Revised)	Human (Annotators iteratively revised the original text)
LOGICNLG (Chen et al., 2020)	37K	Wikipedia (open-domain)	Annotator Generator	Human/Automated (Based on TabFact; annotators wrote “re-fute”, “entailment” statements)
SciGEN (LARGE) (Moosavi et al., 2021)	53K	Scientific	arXiv (scientific articles)	Expert/Automated (Test data annotated by the article’s authors)
NUMERICNLG (Suadaa et al., 2021)	1.3K	Scientific	ACL Anthology (scientific papers)	Expert/Automated (CS experts cleaned and annotated descriptions)
CTRLSciTAB (Guo et al., 2024)	9.0K	Scientific	arXiv (scientific articles)	Domain experts refined and annotated highlighted cells
ROTWIRE (Wiseman et al., 2017)	11K	Basketball	RotoWire	Automated
WEBNLG (Gardent et al., 2017)	2.53K	15 DBPedia categories	Annotator Generator	Crowd workers wrote natural text expressing triple sets
E2E (Novikova et al., 2017)	51K	Restaurants	Annotator Generator	Crowd workers verbalized MR info, could skip unimportant ones
SciTABLES(OURS)	119K	Scientific	arXiv (scientific articles)	Automated

Table 1: Overview of Table-to-Text Datasets. “Pairs” indicates the number of structured data-description pairs (Table, Corresponding paragraph) in each dataset.

Domain (CS Category)	Articles	Pairs	Cap. Sent	Cap. Word	Desc. Sent	Desc. Word
Machine Learning (LG)	22013	42711	1.84	30.07	4.05	88.42
Computer Vision(CV)	18204	38424	1.98	31.31	4.31	90.20
Computation and Language (CL)	12234	27131	1.88	30.25	3.95	86.26
Artificial Intelligence (AI)	11140	23084	1.82	28.68	4.05	87.94
Cryptography and Security (CR)	3303	6597	1.51	22.18	4.04	88.55
Robotics (RO)	2777	4571	1.63	24.26	4.18	90.72
Sound (SD)	2293	4247	1.70	26.36	4.25	92.42
Information Retrieval (IR)	2190	4108	1.63	24.47	4.12	89.90
Software Engineering (SE)	1781	3787	1.33	18.09	4.09	89.64
Social and Information Networks (SI)	1727	3163	1.65	27.68	4.11	92.76
Other	17946	31592	1.54	25.45	3.82	86.59

Table 2: The statistic of the data in Sci-Tables. All domains consist of the “Computer Science” articles from arXiv.org. The Article and Pairs columns show the number of articles and tables in each domain, respectively. The Cap. sent and Cap. word columns show the average number of containing sentences and words in captions, respectively. Similarly, Desc. sent and Desc. word report the average number of sentences and words that are included in the descriptions

addresses these limitations by leveraging naturally occurring, expert-authored scientific descriptions.

3 Dataset Construction

3.1 Source and Extraction

We construct a domain-specific dataset **SciTables** for table-to-text generation using scientific articles from the Computer Science section of arXiv, spanning the years 2017 to 2023. The dataset consists of structured tables paired with their corresponding descriptive paragraphs, extracted from a diverse set of subfields. Figure 2 illustrates the distribution of these papers across Computer Science domains. Our focus is on capturing the complexity and richness of real-world scientific tables. These tables frequently contain dense numeric, symbolic, and categorical data, and are typically accompanied by highly technical textual descriptions authored by domain experts. To collect this data, we extract tables and associated text from LaTeX-formatted source files available in the arXiv repository, ensuring the preservation of structure and semantics. Table 3 provides a statistical summary of the dataset

Year	LaTeX Files	Original	Step 1	Step 2
2017	30,813	18,174	5,504	5,380
2018	41,950	27,848	8,627	8,448
2019	55,525	42,869	13,090	12,830
2020	71,425	32,286	9,866	9,655
2021	77,519	73,565	22,583	22,151
2022	82,109	84,176	26,056	25,537
2023	100,020	112,194	35,591	34,894
Total	459,361	391,112	121,317	118,895

Table 3: Statistics of our dataset construction. **LaTeX Files** is the number of files parsed; **Original** refers to table-reference pairs extracted. Step 1 removes entries with multiple paragraph references; Step 2 filters out descriptions with 100–500 tokens.

construction process, including the number of entries retained at each filtering stage by publication year.

3.2 Automatic Preprocessing and Cleaning

To analyze scientific tables in research articles, we began with the arXiv metadata snapshot, which provides bibliographic information (titles, abstracts, and category labels etc.) for millions of preprints submitted to arXiv. We parsed this dataset to ex-

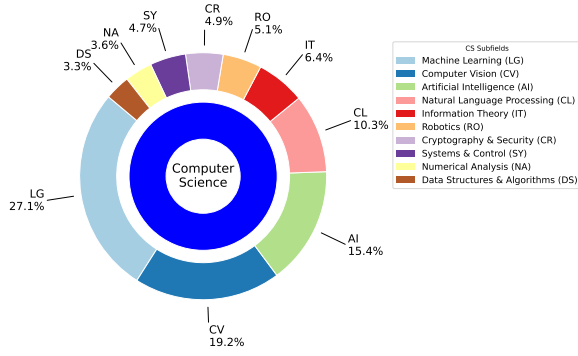


Figure 2: Overview of top 10 CS subfields.

tract high-level category distributions and identify the ten most frequent subcategories within the Computer Science domain (see fig 2). Following meta-data extraction, we conducted a multi-stage pre-processing pipeline to clean, filter, and organize content for downstream modeling. While several paragraphs in a paper may reference the same table, we retain only those that refer exclusively to a single table without simultaneously mentioning other tables or figures. This filtering choice was made to ensure that the model’s focus remains solely on the content of the specific table. We further filtered the paragraphs by length, selecting those between 100 and 500 tokens. This range was chosen to exclude overly short texts lacking sufficient semantic content and overly long ones that may span multiple ideas or references. By focusing on mid-length paragraphs, we aim to maintain both contextual informativeness and alignment with the associated table. Figure 4 shows the token length distribution of the corresponding paragraphs for publications. The pink-highlighted ranges were selected for the next step analysis. The relevant paragraphs extracted from the scientific documents were originally in LaTeX format. Since our goal is to compare these paragraphs with corresponding plain English descriptions in terms of semantic similarity. A crucial step involved converting them into fluent English without LaTeX formatting. To achieve this, we utilized the Llama3.3-70B-Instruct model, which translated technical LaTeX text into coherent, plain English while preserving the original meaning. This conversion step enabled a fair and consistent comparison across formats in the downstream similarity evaluation.

4 Experimental Setup and Evaluation

First, we will benchmark the performance of state-of-the-art large language models (LLMs) on our newly constructed **SciTables** dataset, which is designed to evaluate a model’s ability to perform table-to-text generation in scientific domains. In this task, the input consists of a structured scientific table—often summarizing experimental results, measurements, or comparative analyses and the goal is to generate a coherent, factually accurate, and semantically faithful paragraph that captures the key information and insights conveyed by the table. Our evaluation focuses on three leading instruction-tuned LLMs: *Llama-3.3-70B-Instruct*, *Mistral-Small-24B-Instruct*, and *Gemma-3-27B-Instruct*. These models were selected based on their strong performance across a variety of downstream tasks and their widespread accessibility to the research community. To ensure a comprehensive assessment of model capabilities on tabular scientific data, we will employ a combination of evaluation strategies: (i) automatic metrics that capture semantic similarity between the model-generated paragraph and the ground-truth reference, (ii) an LLM-as-a-judge framework for scalable automated evaluation, and (iii) manual human evaluation focusing on factual correctness and reasoning quality.

4.1 Automatic Metrics

To systematically assess the semantic similarity/variability between the target paragraph generated by LLMs and the corresponding original reference text, we employ several key evaluation metrics. First, we employ a **SBERT Score** (Reimers and Gurevych, 2019) as a metric to assess the semantic alignment between the generated and reference paragraph. Sentence-BERT (SBERT) applies a triplet network structure fine-tune BERT for semantic similarity task. This allows SBERT to generate fixed-size sentence embeddings suitable for cosine similarity computation. However, SBERT still inherits some limitations from underlying transformer models, such as reduced sensitivity to domain-specific nuances or numerical reasoning SBERT Score leverages contextual embeddings from the pre-trained Transformer model BERT (Devlin et al., 2019), RoBERTA (Liu et al., 2019) to compute the similarity at the token-level, capturing deeper semantic relationships beyond the superficial-level overlap. This allows us to

quantify how well a generated paragraph preserves the meaning of the source reference despite potential phrasing variations. In our experiments, we compared two embedding models for computing SBERT similarity, all-MiniLM-L6-v2 and BAAI/bge-large-en-v1.5. To better understand the behavior of these embeddings in capturing semantic similarity, we performed a comparative analysis using both models. The BAAI/bge-large-en-v1.5 model consistently yields higher similarity scores with lower variance, suggesting that it is more robust in maintaining semantic coherence even in the presence of token-level variation. This is also supported by the Pearson correlation matrices, where the correlation between token difference and SBERT score is slightly stronger for bge-large compared to MiniLM. Therefore, although both models effectively capture semantic similarity trends, BAAI/bge-large-en-v1.5 appears to be more sensitive to subtle semantic mismatches and more suitable for our evaluation. As represented in Figure 5 across all models, we observe a moderate to strong negative correlation between SBERT scores and token difference (e.g., -0.29 for Llama, -0.38 for Mistral, and -0.34 for Gemma). This suggests that larger discrepancies in length between generated and reference paragraphs tend to reduce semantic similarity, as measured by SBERT. Interestingly, Gemma and Mistral both show a higher positive correlation between SBERT score and their own output length (0.70), indicating that longer outputs may better preserve semantic alignment in those models. In contrast, Llama shows a weaker correlation (0.58), and all models show negligible correlation between SBERT score and the reference length itself (less than 0.11). These results demonstrate that output verbosity and token-level alignment play a measurable role in semantic fidelity across different LLMs, with Gemma and Mistral benefiting more from increased generation length in terms of semantic similarity.

Table 4 shows a comparative evaluation of LLM-generated text quality across different input table sizes. Across all models, performance declines markedly as table size increases: small tables yield high similarity and quality scores (SBERT ~ 0.78 – 0.80 , BERTScore ~ 0.84 – 0.85), while large tables drop substantially (SBERT ~ 0.46 – 0.49 , BERTScore ~ 0.81). All the models perform comparably overall, but all exhibit the same trend of degradation. As input size grows, KL divergence

rises sharply (from ~ 7 – 8 for small to ~ 16 for large tables), indicating greater content drift between the input table caption and generated outputs when models are overloaded with information. These patterns suggest that while current LLMs can handle concise, information-light inputs effectively, they struggle to retain both fidelity and variety as complexity increases. The combination of declining similarity and higher divergence highlight a key limitation in current models’ ability to summarize or generate faithful text from large, information-rich tables.

4.1.1 Impact of providing captions in Table-to-Text Generation

To better assess the role of captions in table-to-text generation, we masked them from the input tables. When captions are provided (Table 4), all three models—Llama, Mistral, and Gemma—achieve consistently higher similarity scores across SBERT, BERTScore F1, BLEURT, BLEU, ROUGE-L, and METEOR. This shows that models effectively leverage captions to anchor their outputs, producing more accurate and faithful descriptions. For instance, SBERT scores remain in the 0.72 – 0.80 range with captions but drop sharply to ~ 0.48 – 0.51 in the masked setting, underscoring the loss of semantic alignment without explicit caption guidance. In the absence of captions, model outputs become less semantically consistent with the reference paragraphs, and all similarity-based metrics degrade (see Table 5). BLEURT scores are markedly lower (more negative), reflecting reduced text quality and coherence, while overlap-based measures such as ROUGE-L and METEOR also decline, especially for larger tables. This suggests that captions are particularly important for helping models handle longer and more complex table inputs. Interestingly, divergence-based measures (KL_{LC}) increase in the masked caption setting, especially for large tables, highlighting that the lexical distribution of LLM outputs drifts further away from that of the gold captions when no caption anchor is present. This reinforces the idea that captions provide not only semantic context but also lexical grounding, helping models produce text that better mirrors human-authored descriptions.

4.2 LLM-Based Evaluation of Semantic similarity

To evaluate the semantic similarity between reference paragraphs and model-generated sentences,

Model	Table Size	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	KL _{LC}
Llama	Small	0.779 ± 0.078	0.853 ± 0.019	-0.595 ± 0.255	0.021 ± 0.021	0.191 ± 0.057	0.204 ± 0.076	7.976 ± 4.384
	Medium	0.700 ± 0.156	0.841 ± 0.022	-0.804 ± 0.401	0.018 ± 0.020	0.167 ± 0.061	0.181 ± 0.090	10.686 ± 5.454
	Large	0.491 ± 0.060	0.817 ± 0.012	-1.299 ± 0.210	0.005 ± 0.007	0.113 ± 0.039	0.085 ± 0.045	16.444 ± 2.627
	Overall	0.746 ± 0.122	0.848 ± 0.021	-0.680 ± 0.339	0.020 ± 0.020	0.181 ± 0.060	0.193 ± 0.083	9.071 ± 5.032
Mistral	Small	0.796 ± 0.076	0.844 ± 0.020	-0.673 ± 0.255	0.024 ± 0.022	0.194 ± 0.059	0.224 ± 0.073	6.673 ± 4.282
	Medium	0.690 ± 0.156	0.832 ± 0.022	-0.901 ± 0.401	0.017 ± 0.020	0.164 ± 0.061	0.184 ± 0.094	10.306 ± 5.882
	Large	0.464 ± 0.067	0.810 ± 0.014	-1.345 ± 0.210	0.005 ± 0.007	0.115 ± 0.041	0.085 ± 0.044	16.417 ± 2.794
	Overall	0.753 ± 0.135	0.840 ± 0.022	-0.764 ± 0.341	0.021 ± 0.022	0.182 ± 0.062	0.207 ± 0.084	8.101 ± 5.292
Gemma	Small	0.787 ± 0.074	0.850 ± 0.019	-0.648 ± 0.254	0.020 ± 0.018	0.190 ± 0.054	0.211 ± 0.069	8.520 ± 4.160
	Medium	0.688 ± 0.161	0.837 ± 0.021	-0.883 ± 0.384	0.015 ± 0.017	0.161 ± 0.059	0.173 ± 0.087	11.415 ± 5.255
	Large	0.492 ± 0.064	0.817 ± 0.013	-1.296 ± 0.223	0.006 ± 0.008	0.113 ± 0.040	0.085 ± 0.044	16.429 ± 2.676
	Overall	0.747 ± 0.126	0.845 ± 0.020	-0.741 ± 0.333	0.018 ± 0.018	0.179 ± 0.058	0.195 ± 0.079	9.662 ± 4.838

Table 4: Evaluation of LLM-generated text against reference paragraphs grouped by table size (Small: <500 tokens, Medium: 500–2000, Large: >2000, Overall). KL_{LC} is the KL divergence between the unigram distribution of the caption and LLM generated text.

Model	Table Size	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	KL _{LC}
Llama	Small	0.711 ± 0.084	0.844 ± 0.017	-0.720 ± 0.233	0.017 ± 0.015	0.175 ± 0.057	0.197 ± 0.061	12.478 ± 3.850
	Medium	0.681 ± 0.126	0.838 ± 0.019	-0.842 ± 0.359	0.014 ± 0.014	0.158 ± 0.061	0.176 ± 0.078	13.371 ± 3.981
	Large	0.506 ± 0.068	0.818 ± 0.012	-1.334 ± 0.209	0.005 ± 0.007	0.108 ± 0.039	0.079 ± 0.044	17.344 ± 2.295
	Overall	0.696 ± 0.105	0.841 ± 0.018	-0.775 ± 0.299	0.016 ± 0.015	0.168 ± 0.052	0.187 ± 0.069	12.888 ± 3.948
Mistral	Small	0.778 ± 0.076	0.843 ± 0.020	-0.682 ± 0.255	0.023 ± 0.050	0.191 ± 0.059	0.221 ± 0.072	7.419 ± 4.698
	Medium	0.695 ± 0.156	0.834 ± 0.021	-0.894 ± 0.411	0.017 ± 0.054	0.163 ± 0.061	0.184 ± 0.094	10.704 ± 5.880
	Large	0.488 ± 0.067	0.815 ± 0.014	-1.380 ± 0.209	0.005 ± 0.007	0.111 ± 0.041	0.081 ± 0.049	17.252 ± 2.598
	Overall	0.744 ± 0.131	0.840 ± 0.021	-0.768 ± 0.344	0.021 ± 0.022	0.180 ± 0.062	0.206 ± 0.083	8.733 ± 5.471
Gemma	Small	0.739 ± 0.084	0.844 ± 0.019	-0.673 ± 0.255	0.017 ± 0.059	0.178 ± 0.051	0.195 ± 0.063	12.389 ± 3.831
	Medium	0.686 ± 0.137	0.835 ± 0.021	-0.869 ± 0.411	0.013 ± 0.061	0.155 ± 0.055	0.168 ± 0.078	13.517 ± 4.107
	Large	0.505 ± 0.067	0.818 ± 0.013	-1.334 ± 0.264	0.005 ± 0.007	0.109 ± 0.042	0.077 ± 0.038	17.401 ± 2.294
	Overall	0.716 ± 0.112	0.841 ± 0.019	-0.753 ± 0.319	0.015 ± 0.014	0.169 ± 0.054	0.183 ± 0.071	12.880 ± 3.994

Table 5: Masked caption: Evaluation of LLM-generated text against reference paragraphs grouped by table size (Small: <500 tokens, Medium: 500–2000, Large: >2000, Overall) with standard deviations. KL_{LC} is the KL divergence between caption and LLM output unigram distributions.

we employed a separate LLM as an automatic judge. Prior research (Gilardi et al., 2023; Huang et al., 2023) has shown that LLMs can serve as reliable evaluators, offering both scalability and interpretability (Zheng et al., 2023). At the same time, relying on LLMs for evaluation introduces challenges such as position bias, which has been noted in both human decision-making (Blunch, 1984) and LLM behavior (Ko et al., 2020). In this study, we used Qwen-2.5-72B, a 72-billion-parameter instruction-tuned language model, as the automatic judge. We used the rubrics (see Appendix) to evaluate semantic similarity between reference paragraphs and model-generated sentences.

To validate the reliability of the LLM-based evaluation, we conducted a human evaluation involving three annotators, each assigned a distinct subset of the dataset. Annotators independently rated the semantic similarity between generated sentences (produced by three different LLMs) and the corresponding reference paragraphs. We then calculated the Pearson correlation Coefficient between the hu-

man rating and LLM-judge’s ratings for each model across 200 random samples. This allowed us to assess the degree of alignment between human and LLM-based evaluations, as well as examine the impact of annotator subjectivity. Figure 3 presents a radar chart illustrating the correlations. Overall, the results show moderate to strong correlations across all models and annotators, supporting the reliability of the LLM-based evaluation method. Table 6 summarizes the comparison between human and LLM judgments. Overall, LLM-judge assigns slightly higher similarity scores than human annotators but follows the same general trends. Correlations between human and LLM-judge ratings are consistently positive and statistically significant across all models and annotator segments. Notably, Mistral shows the strongest alignment (overall Pearson $r = 0.58$, with Annotator 3 reaching 0.85), while Gemma and Llama also exhibit meaningful correlations (~ 0.51 – 0.52 overall). These findings suggest that the LLM-judge captures human semantic similarity assessments reasonably well, though the

Annotator Segment	Model	Human Mean $\pm$ Std	Qwen Mean $\pm$ Std	Pearson r	p-value
Annotator 1 (R:1–70)	Llama	4.07 $\pm$ 0.52	4.36 $\pm$ 0.48	0.59	7.6e-8
Annotator 1 (R:1–70)	Mistral	4.23 $\pm$ 0.67	4.59 $\pm$ 0.49	0.51	5.82e-6
Annotator 1 (R:1–70)	Gemma	4.27 $\pm$ 0.61	4.48 $\pm$ 0.50	0.47	0.000105
Annotator 2 (R:71–140)	Llama	4.17 $\pm$ 0.74	4.45 $\pm$ 0.53	0.61	6.25e-9
Annotator 2 (R:71–140)	Mistral	4.38 $\pm$ 0.78	4.7 $\pm$ 0.49	0.53	5.87e-7
Annotator 2(R:71–140)	Gemma	4.43 $\pm$ 0.58	4.52 $\pm$ 0.50	0.60	1.08e-6
Annotator 3 (R:141–200)	Llama	4.45 $\pm$ 0.56	4.21 $\pm$ 0.49	0.56	4.34e-5
Annotator 3 (R:141–200)	Mistral	4.55 $\pm$ 0.53	4.6 $\pm$ 0.53	0.85	3.8e-16
Annotator 3 (R:141–200)	Gemma	4.48 $\pm$ 0.53	4.33 $\pm$ 0.51	0.52	4.04e-7
Overall (R:1–200)	Llama	4.22 $\pm$ 0.63	4.35 $\pm$ 0.51	0.52	2.81e-15
Overall (R:1–200)	Mistral	4.38 $\pm$ 0.68	4.63 $\pm$ 0.50	0.58	1.69e-19
Overall (R:1–200)	Gemma	4.39 $\pm$ 0.58	4.45 $\pm$ 0.51	0.51	3.25e-14

Table 6: Comparison of human and LLM-judge (Qwen-2.5-72B) similarity scores for model-generated text with reference paragraphs. The table reports mean $\pm$ standard deviation of similarity ratings, along with Pearson correlation coefficients and statistical significance (p-values) across annotator segments and models.

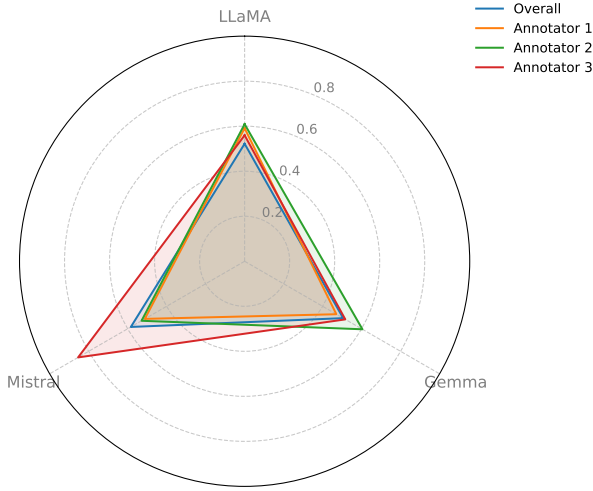


Figure 3: Pearson Correlation between Human and LLM-judge.

strength of agreement varies by model and annotator. Overall, the results highlight the promise of high-capacity LLMs as approximate evaluators of semantic similarity while underscoring the role of annotator subjectivity in nuanced judgment tasks.

4.3 Other Evaluation Metrics

While semantic similarity provides an overall indication of content alignment between generated text and reference paragraphs, it does not capture deeper aspects of output quality. A model may produce a semantically similar sentence that is factually inaccurate, logically shallow, or includes irrelevant information. To achieve a holistic assessment of model behavior, we therefore incorporate four complementary evaluation dimensions—factual correctness, reasoning depth, rea-

soning type, and relevance—each targeting distinct aspects of reasoning quality and interpretive alignment.

- **Factual Correctness** measures whether the generated text accurately reflects the facts presented in the source table or reference paragraph. High factuality indicates that the model grounds its generation in the provided evidence rather than hallucinating unsupported claims.
- **Reasoning Depth** evaluates the level of inferential complexity in the response. A higher score denotes the ability to draw non-trivial or multi-step conclusions beyond surface-level reporting.
- **Reasoning Type** characterizes the nature of reasoning applied—numerical, comparative, temporal, commonsense, or none—providing insight into the cognitive strategies employed by the model.
- **Relevance** captures how well the generated content stays focused on the pertinent information, excluding redundant or tangential material.

We employ Qwen-2.5-72B as the LLM-judge to evaluate the key output-quality dimensions—factual correctness, reasoning depth, reasoning type, and relevance—for table-to-text generation outputs produced by the models - Llama, Mistral, and Gemma. To assess the reliability of using LLM as an automatic evaluator, we compared the Qwen-2.5-72B model’s judgments with those of human annotators across these dimensions.

As shown in Table 8, Gemma and Mistral’s generated descriptions show statistically significant alignment between LLM-judge (Qwen) and human evaluators across all metrics ($p < 0.05$), with moderate correlations ($\phi \sim 0.36\text{--}0.50$, $\kappa \sim 0.38\text{--}0.55$). This suggests that Qwen’s evaluations broadly mirror human judgments for these models. Across all models, Qwen’s factual and relevance assessments closely track human annotations. For instance, factual accuracy remains high for all systems. Similarly, relevance scores are above 0.84 across all cases (see Table 7). These findings indicate that the LLM-judge effectively identifies whether generated text remains faithful to source data and focused on the intended content, making it a reliable proxy for surface-level quality judgments. Further, across all models, the reasoning depth of generated responses is consistently low, indicating that the LLMs largely fail to provide depth in reasoning. Additionally, Qwen as LLM-judge underestimates reasoning depth, assigning lower scores than humans and showing only moderate agreement. A clear divergence emerges in reasoning type judgments: Qwen labels a substantially higher proportion of outputs as “no reasoning” (e.g., 82.6% for Mistral vs. 63.0% by humans) and fewer as “comparative” or “analytical.” This indicates that while Qwen reliably detects explicit reasoning cues (e.g., numerical comparisons), it tends to overlook subtler forms of implicit inference recognized by human evaluators. These results further demonstrate that LLM-based judges can effectively complement human evaluation for large-scale benchmarking, but human oversight remains essential for assessing inferential and reasoning-rich content.

5 Conclusions

We introduce **SciTables**, a large benchmark dataset for scientific table-to-text generation, featuring complex numeric and symbolic tables paired with naturally occurring textual descriptions from arXiv CS papers. We evaluated several state-of-the-art LLMs across multiple metrics, including semantic similarity, factual correctness, reasoning depth, relevance, and reasoning type. While model outputs achieve high semantic alignment with reference paragraphs, they consistently lack reasoning depth. Our evaluation leverages both automatic metrics and the LLM-as-judge framework, validated through human assessments, highlighting that LLM-based judges can complement human

Models	Metrics	Human	LLM-judge
Llama	Factual-correctness	97.78%	93.33%
	Reasoning depth	0.33	0.13
	Reasoning type	No Reasoning	66.67%
		Comparative	33.33%
	Relevance	0.98	0.93
Mistral	Factual-correctness	86.96%	84.78%
	Reasoning depth	0.37	0.20
	Reasoning type	No Reasoning	63.04%
		Comparative	36.96%
	Relevance	0.87	0.85
Gemma	Factual-correctness	86.67%	84.44%
	Reasoning depth	0.42	0.29
	Reasoning type	No Reasoning	57.78%
		Comparative	42.22%
	Relevance	0.87	0.84

Table 7: Comparison between human and LLM-as-judge (Qwen-2.5-72B) evaluations of LLM-generated outputs across key quality metrics, including factual correctness, reasoning depth, relevance, and reasoning type distribution.

Model	Metric	χ^2	p-value	Φ	Cohen’s κ
Mistral24B	Factual	9.94	0.0016*	0.46	0.55
	Reasoning Depth	5.97	0.0145*	0.36	0.38
	Reasoning Type	8.15	0.0043*	0.42	0.42
	Relevance	9.94	0.0016*	0.46	0.55
Llama-3.3-70B	Factual	3.09	0.0789	0.26	0.48
	Reasoning Depth	1.95	0.1629	0.21	0.24
	Reasoning Type	3.40	0.0651	0.28	0.28
	Relevance	3.09	0.0789	0.26	0.48
Gemma	Factual	9.64	0.0019*	0.46	0.55
	Reasoning Depth	11.13	0.0008*	0.50	0.52
	Reasoning Type	8.95	0.0028*	0.45	0.48
	Relevance	9.64	0.0019*	0.46	0.55

Table 8: χ^2 , Φ coefficient, p-value, and Cohen’s κ for agreement between human annotators and the LLM-judge (Qwen-2.5-72B) in evaluating various LLM-generated outputs across different output quality metrics. Statistically significant results ($p < 0.05$) are marked with an asterisk (*).

evaluation for large-scale benchmarking, particularly for factual and relevance-oriented assessments. **SciTables** provides a realistic and scalable framework to advance faithful, coherent, and informative table-to-text generation in scientific domains.

6 Ethical Considerations

The ethical considerations surrounding the generation of description from structured table encompass several key aspects:

- **Accuracy and Fidelity:** Ensuring that the generated description truthfully represents the underlying table without introducing errors or misleading claims.
- **Bias and Fairness:** Preventing the reinforcement or amplification of biases that may be embedded in the data or learned by the model.
- **Transparency and Accountability:** Providing clarity about data sources and generation processes so that outputs can be traced, verified, and responsibly used.

7 Limitations

Here are some limitations of this work:

- While SBERT similarity scores provide a scalable automatic measure of semantic alignment, they inherit limitations from the underlying transformer architecture. In particular, they may exhibit reduced sensitivity to domain-specific nuances or numerical reasoning, potentially affecting performance on highly technical or tabular content.
- The corresponding text retrieved for each table represents the only paragraph referencing that table; however, it may not provide a complete explanation of the table's contents.
- This study focused primarily on numerical tables in the CS domain. Future work could extend the evaluation to text-rich tables in other domains to test generalizability.
- Fine-tuning LLMs on this extracted dataset has the potential to further enhance model performance, particularly in table-to-text generation tasks.

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A Appendix

A.1 Prompts

Prompt for LaTeX-to-Plain Text Conversion

Make the following paragraph LaTeX command free and convert it to plain English text. Keep everything else the same. Do not do paraphrasing. Add the `[END]` token at the end of the cleaned paragraph.

Paragraph: {referencing_paragraphs}

Answer:

Prompt for Table-Based Text Generation (With Caption)

Generate a concise and informative sentence that summarizes the key information from the given table and its caption. You may use the caption as additional context. Add the `[END]` token at the end.

Caption: {caption}

Table: {table}

Answer:

Prompt for Semantic Similarity Judgment Between Generated Sentence and Reference Paragraph

You are an evaluator tasked with identifying semantic similarities between two descriptions of the same table. Compare the `**meaning**` of both texts, not their exact words. Rate their semantic similarity on a scale from 1 to 5, where:

- 1 = Completely different meanings
- 2 = Slight overlap in meaning
- 3 = Somewhat similar meaning
- 4 = Very similar meaning
- 5 = Nearly identical in meaning

Only use the information provided. Do not rely on external knowledge. Add a brief justification (1-2 sentences). Add the `[END]` token at the end.

Table: {table}

Reference Paragraph: Reference_paragraph

Generated Sentence: Generated_sentence

Answer:

A.2 Table Size Distribution

To investigate whether table size impacts model performance, we grouped tables based on the number of tokens in their table-content fields which are in LaTeX format, using the tiktoken tokenizer with the encoding scheme of gpt-3.5-turbo. Specifically, we defined three size categories: Small (fewer than 500

Token Range	Count	Percentage (%)
<100	60	0.67
100–250	2112	23.71
250–500	3510	39.41
500–1K	2217	24.89
1K–2K	782	8.78
2K–5K	206	2.31
5K–10K	19	0.21

Table 9: Distribution of table sizes based on token counts.

tokens), Medium (500 to 2000 tokens), and Large (more than 2000 tokens). The detailed distribution of table sizes can be found in Table 9. This binning allows us to analyze whether language models perform differently depending on the amount of structured data they receive. For each group, we report the average performance across multiple evaluation metrics, including LLM-as-Judge scores, SBERT similarity, BLEU, and ROUGE-L.

A.3 LLM-generated descriptions and original paragraphs, grouped by reference paragraph word count

In Table 10, we show the similarity between the LLM-generated descriptions and the reference paragraphs, grouped by the reference paragraph word counts. The results show that the similarity increases with increase in the reference paragraph size. Table 11, we show the similarity between the LLM-generated descriptions and the reference paragraphs, grouped by the word counts of the reference paragraphs where no table captions are provided. The results show that similarity increases with increase in the reference paragraph size but overall similarity decreases due to lack of caption information.

A.4 Similarity with reference paragraph and Captions

To better understand whether caption influence the generated text, we computed similarity between the LLM-generated descriptions and the table captions. Table 12 shows the similarity between the LLM-generated descriptions and the captions, grouped by the table size where as Table 13 shows the similarity between the LLM-generated descriptions and the captions, grouped by the table caption word counts.

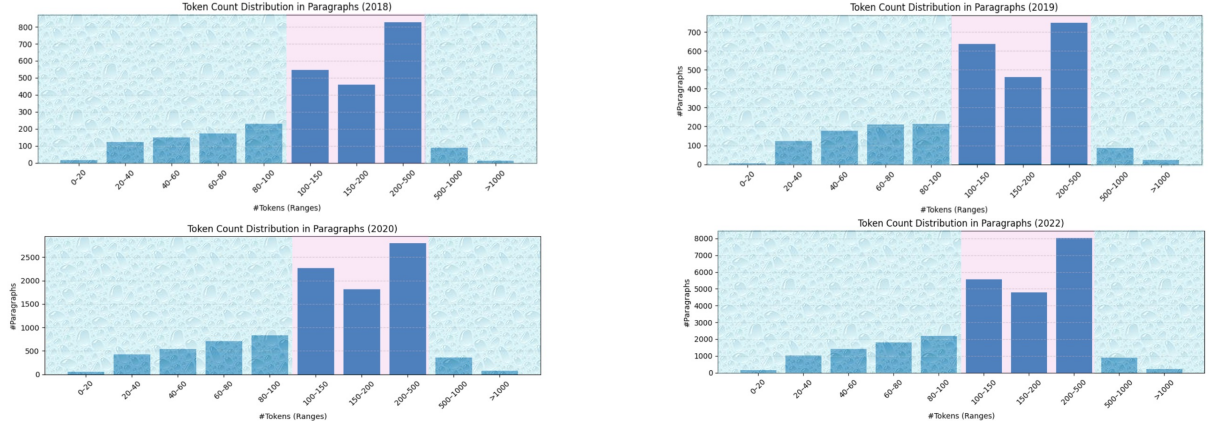


Figure 4: Token count distributions of the reference paragraphs associated with tables from different publication years (2018, 2019, 2020, and 2022).

Model	Ref Word Count	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	$\Delta\text{Simpson}_{LC}$	$\Delta\text{Simpson}_{LO}$	KL_{LC}
Llama	Small	0.725 ± 0.117	0.851 ± 0.023	-0.678 ± 0.339	0.022 ± 0.021	0.182 ± 0.021	0.231 ± 0.098	0.092 ± 0.307	0.963 ± 0.018	9.230 ± 5.265
	Medium	0.755 ± 0.123	0.849 ± 0.020	-0.676 ± 0.339	0.022 ± 0.020	0.187 ± 0.022	0.195 ± 0.075	0.078 ± 0.347	0.964 ± 0.013	8.998 ± 4.947
	Large	0.751 ± 0.123	0.844 ± 0.019	-0.690 ± 0.339	0.013 ± 0.007	0.169 ± 0.014	0.154 ± 0.060	0.080 ± 0.359	0.965 ± 0.014	9.044 ± 4.947
	Overall	0.746 ± 0.122	0.848 ± 0.021	-0.680 ± 0.339	0.020 ± 0.020	0.181 ± 0.060	0.193 ± 0.083	0.082 ± 0.137	0.964 ± 0.015	9.071 ± 5.032
Mistral	Small	0.724 ± 0.135	0.839 ± 0.023	-0.764 ± 0.341	0.019 ± 0.022	0.167 ± 0.018	0.228 ± 0.093	0.093 ± 0.149	0.964 ± 0.019	8.355 ± 5.534
	Medium	0.763 ± 0.135	0.841 ± 0.021	-0.764 ± 0.341	0.024 ± 0.022	0.191 ± 0.024	0.214 ± 0.082	0.078 ± 0.128	0.965 ± 0.014	7.992 ± 5.190
	Large	0.764 ± 0.135	0.838 ± 0.022	-0.764 ± 0.341	0.019 ± 0.022	0.180 ± 0.019	0.177 ± 0.068	0.081 ± 0.142	0.965 ± 0.015	8.046 ± 5.220
	Overall	0.753 ± 0.135	0.840 ± 0.022	-0.764 ± 0.341	0.021 ± 0.022	0.182 ± 0.062	0.207 ± 0.084	0.083 ± 0.138	0.965 ± 0.016	8.101 ± 5.292
Gemma	Small	0.720 ± 0.126	0.850 ± 0.019	-0.648 ± 0.254	0.020 ± 0.018	0.190 ± 0.054	0.211 ± 0.069	0.094 ± 0.149	0.964 ± 0.019	9.800 ± 5.077
	Medium	0.757 ± 0.126	0.837 ± 0.021	-0.883 ± 0.384	0.015 ± 0.017	0.161 ± 0.059	0.173 ± 0.087	0.079 ± 0.128	0.966 ± 0.014	9.584 ± 4.755
	Large	0.755 ± 0.126	0.817 ± 0.013	-1.296 ± 0.223	0.006 ± 0.008	0.113 ± 0.040	0.085 ± 0.044	0.082 ± 0.142	0.966 ± 0.015	9.664 ± 4.742
	Overall	0.747 ± 0.126	0.845 ± 0.020	-0.741 ± 0.333	0.018 ± 0.018	0.179 ± 0.058	0.195 ± 0.079	0.084 ± 0.138	0.966 ± 0.015	9.662 ± 4.838

Table 10: Similarity between LLM-generated descriptions and original reference paragraphs, grouped by reference paragraph word count. $\Delta\text{Simpson}_{LC}$ denote lexical diversity differences between LLM generated description and table caption while $\Delta\text{Simpson}_{LO}$ denote lexical diversity differences between LLM generated description and original reference paragraph. KL_{LC} is the KL divergence between the unigram distribution of the caption and LLM generated text.

B Annotator Demographics and Evaluation Procedure:

Three authors were chosen as annotators for human-based evaluations. The annotators were provided with the same template with rubrics that was used for the LLM-based evaluation.

C AI assistance in Writing

AI Assistants were used solely to assist in improving the writing clarity and language of this paper. Specifically, AI-assisted refinements were applied to enhance readability, coherence, and grammatical accuracy. No AI-generated content was used to replace critical thinking or fabricate results. Ideas, methodology, experimental design, analysis, and conclusions were entirely conceived, developed, and executed by the authors.

Model	RefP word-count	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	$\Delta\text{Simpson}_{LC}$	$\Delta\text{Simpson}_{LO}$	KL _{LC}
Llama	Small	0.682 $\pm$ 0.100	0.843 $\pm$ 0.020	-0.783 $\pm$ 0.299	0.015 $\pm$ 0.013	0.158 $\pm$ 0.014	0.209 $\pm$ 0.078	0.094 $\pm$ 0.146	0.966 $\pm$ 0.014	12.951 $\pm$ 4.121
	Medium	0.704 $\pm$ 0.106	0.843 $\pm$ 0.018	-0.760 $\pm$ 0.282	0.018 $\pm$ 0.016	0.176 $\pm$ 0.016	0.192 $\pm$ 0.066	0.080 $\pm$ 0.127	0.967 $\pm$ 0.013	12.861 $\pm$ 3.909
	Large	0.697 $\pm$ 0.106	0.838 $\pm$ 0.017	-0.791 $\pm$ 0.311	0.013 $\pm$ 0.011	0.164 $\pm$ 0.013	0.159 $\pm$ 0.057	0.079 $\pm$ 0.136	0.967 $\pm$ 0.013	12.875 $\pm$ 3.846
	Overall	0.696 $\pm$ 0.105	0.841 $\pm$ 0.018	-0.775 $\pm$ 0.299	0.016 $\pm$ 0.015	0.168 $\pm$ 0.052	0.187 $\pm$ 0.069	0.083 $\pm$ 0.135	0.967 $\pm$ 0.013	12.888 $\pm$ 3.948
Mistral	Small	0.717 $\pm$ 0.105	0.839 $\pm$ 0.021	-0.779 $\pm$ 0.304	0.018 $\pm$ 0.013	0.165 $\pm$ 0.018	0.226 $\pm$ 0.078	0.091 $\pm$ 0.149	0.963 $\pm$ 0.016	8.948 $\pm$ 5.648
	Medium	0.752 $\pm$ 0.113	0.841 $\pm$ 0.020	-0.757 $\pm$ 0.352	0.023 $\pm$ 0.016	0.190 $\pm$ 0.024	0.211 $\pm$ 0.066	0.077 $\pm$ 0.128	0.964 $\pm$ 0.015	8.603 $\pm$ 5.414
	Large	0.756 $\pm$ 0.115	0.838 $\pm$ 0.020	-0.776 $\pm$ 0.365	0.019 $\pm$ 0.011	0.178 $\pm$ 0.020	0.177 $\pm$ 0.057	0.077 $\pm$ 0.142	0.964 $\pm$ 0.016	8.746 $\pm$ 5.392
	Overall	0.744 $\pm$ 0.131	0.840 $\pm$ 0.021	-0.768 $\pm$ 0.344	0.021 $\pm$ 0.022	0.180 $\pm$ 0.062	0.206 $\pm$ 0.083	0.080 $\pm$ 0.135	0.964 $\pm$ 0.016	8.733 $\pm$ 5.471
Gemma	Small	0.696 $\pm$ 0.121	0.842 $\pm$ 0.019	-0.761 $\pm$ 0.299	0.016 $\pm$ 0.013	0.162 $\pm$ 0.017	0.208 $\pm$ 0.092	0.092 $\pm$ 0.147	0.965 $\pm$ 0.014	12.933 $\pm$ 4.173
	Medium	0.724 $\pm$ 0.132	0.842 $\pm$ 0.021	-0.739 $\pm$ 0.322	0.018 $\pm$ 0.016	0.177 $\pm$ 0.020	0.188 $\pm$ 0.082	0.078 $\pm$ 0.127	0.965 $\pm$ 0.014	12.832 $\pm$ 3.959
	Large	0.721 $\pm$ 0.134	0.838 $\pm$ 0.020	-0.769 $\pm$ 0.330	0.011 $\pm$ 0.011	0.161 $\pm$ 0.013	0.151 $\pm$ 0.070	0.078 $\pm$ 0.137	0.966 $\pm$ 0.014	12.910 $\pm$ 3.876
	Overall	0.716 $\pm$ 0.112	0.841 $\pm$ 0.019	-0.753 $\pm$ 0.319	0.015 $\pm$ 0.014	0.169 $\pm$ 0.054	0.183 $\pm$ 0.071	0.082 $\pm$ 0.135	0.965 $\pm$ 0.014	12.880 $\pm$ 3.994

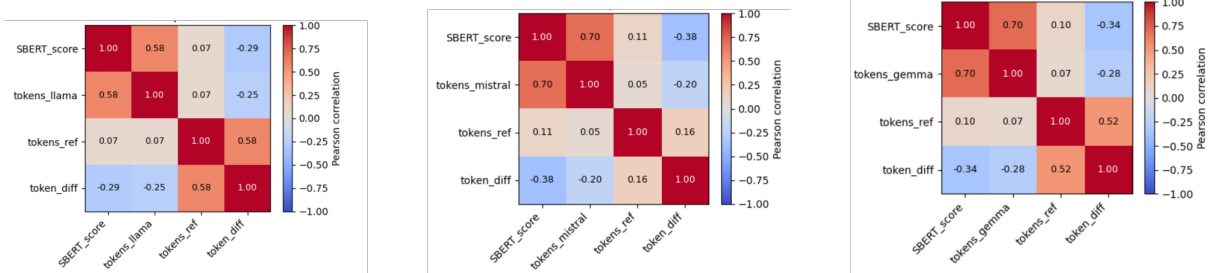
Table 11: Masked caption: Average similarity between LLM-generated descriptions and original reference paragraphs, grouped by reference paragraph word count. $\Delta\text{Simpson}_{LC}$ denote lexical diversity differences between LLM generated description and table caption while $\Delta\text{Simpson}_{LO}$ denote lexical diversity differences between LLM generated description and original reference paragraph. KL_{LC} is the KL divergence between the unigram distribution of the caption and LLM generated text.

Model	Table Size	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	$\Delta\text{Simpson}_{LC}$	$\Delta\text{Simpson}_{LO}$
Llama	Small	0.778 $\pm$ 0.096	0.790 $\pm$ 0.009	-1.726 $\pm$ 0.219	0.000 $\pm$ 0.000	0.003 $\pm$ 0.010	0.006 $\pm$ 0.016	0.092 $\pm$ 0.131	0.966 $\pm$ 0.008
	Medium	0.699 $\pm$ 0.162	0.801 $\pm$ 0.023	-1.736 $\pm$ 0.193	0.000 $\pm$ 0.000	0.005 $\pm$ 0.017	0.008 $\pm$ 0.026	0.062 $\pm$ 0.138	0.962 $\pm$ 0.014
	Large	0.501 $\pm$ 0.071	0.833 $\pm$ 0.013	-1.841 $\pm$ 0.154	0.000 $\pm$ 0.000	0.013 $\pm$ 0.030	0.020 $\pm$ 0.047	0.033 $\pm$ 0.140	0.944 $\pm$ 0.007
	Overall	0.745 $\pm$ 0.133	0.795 $\pm$ 0.018	-1.732 $\pm$ 0.210	0.000 $\pm$ 0.000	0.004 $\pm$ 0.013	0.007 $\pm$ 0.023	0.080 $\pm$ 0.135	0.964 $\pm$ 0.011
Mistral	Small	0.792 $\pm$ 0.000	0.779 $\pm$ 0.009	-1.647 $\pm$ 0.000	0.000 $\pm$ 0.000	0.003 $\pm$ 0.008	0.006 $\pm$ 0.019	0.093 $\pm$ 0.132	0.968 $\pm$ 0.008
	Medium	0.688 $\pm$ 0.000	0.782 $\pm$ 0.010	-1.697 $\pm$ 0.000	0.000 $\pm$ 0.000	0.004 $\pm$ 0.015	0.007 $\pm$ 0.027	0.062 $\pm$ 0.138	0.961 $\pm$ 0.016
	Large	0.474 $\pm$ 0.000	0.792 $\pm$ 0.006	-1.661 $\pm$ 0.000	0.000 $\pm$ 0.000	0.010 $\pm$ 0.028	0.016 $\pm$ 0.046	0.030 $\pm$ 0.140	0.941 $\pm$ 0.008
	Overall	0.748 $\pm$ 0.148	0.780 $\pm$ 0.010	-1.665 $\pm$ 0.221	0.000 $\pm$ 0.000	0.003 $\pm$ 0.012	0.006 $\pm$ 0.021	0.081 $\pm$ 0.135	0.965 $\pm$ 0.013
Gemma	Small	0.770 $\pm$ 0.094	0.786 $\pm$ 0.009	-1.727 $\pm$ 0.212	0.000 $\pm$ 0.000	0.003 $\pm$ 0.010	0.006 $\pm$ 0.018	0.094 $\pm$ 0.132	0.969 $\pm$ 0.007
	Medium	0.680 $\pm$ 0.162	0.801 $\pm$ 0.026	-1.786 $\pm$ 0.171	0.000 $\pm$ 0.000	0.005 $\pm$ 0.018	0.009 $\pm$ 0.028	0.062 $\pm$ 0.138	0.961 $\pm$ 0.015
	Large	0.503 $\pm$ 0.074	0.832 $\pm$ 0.014	-1.848 $\pm$ 0.138	0.000 $\pm$ 0.000	0.013 $\pm$ 0.030	0.020 $\pm$ 0.046	0.034 $\pm$ 0.140	0.944 $\pm$ 0.008
	Overall	0.733 $\pm$ 0.134	0.792 $\pm$ 0.020	-1.750 $\pm$ 0.200	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.007 $\pm$ 0.023	0.082 $\pm$ 0.135	0.966 $\pm$ 0.011

Table 12: Average Sim(LLMGen, Caption) for various similarity and evaluation metrics by table size (Small, Medium, Large, and Overall). SBERT: Sentence-BERT cosine similarity. $\Delta\text{Simpson}_{LC}$ shows lexical diversity difference between LLM output and caption; $\Delta\text{Simpson}_{LO}$ shows difference with the original reference.

Model	RefP word-count	SBERT	BERTScore F1	BLEURT	BLEU	ROUGE-L	METEOR	$\Delta\text{Simpson}_{LC}$	$\Delta\text{Simpson}_{LO}$
Llama	Small	0.740 $\pm$ 0.135	0.796 $\pm$ 0.018	-1.732 $\pm$ 0.208	0.000 $\pm$ 0.000	0.003 $\pm$ 0.013	0.007 $\pm$ 0.023	0.083 $\pm$ 0.139	0.964 $\pm$ 0.012
	Medium	0.749 $\pm$ 0.132	0.794 $\pm$ 0.018	-1.735 $\pm$ 0.208	0.000 $\pm$ 0.000	0.004 $\pm$ 0.013	0.007 $\pm$ 0.023	0.077 $\pm$ 0.127	0.965 $\pm$ 0.011
	Large	0.746 $\pm$ 0.132	0.795 $\pm$ 0.018	-1.727 $\pm$ 0.215	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.008 $\pm$ 0.024	0.079 $\pm$ 0.141	0.965 $\pm$ 0.011
	Overall	0.745 $\pm$ 0.133	0.795 $\pm$ 0.018	-1.732 $\pm$ 0.210	0.000 $\pm$ 0.000	0.004 $\pm$ 0.013	0.007 $\pm$ 0.023	0.080 $\pm$ 0.135	0.964 $\pm$ 0.011
Mistral	Small	0.742 $\pm$ 0.148	0.780 $\pm$ 0.010	-1.664 $\pm$ 0.216	0.000 $\pm$ 0.000	0.003 $\pm$ 0.012	0.006 $\pm$ 0.021	0.084 $\pm$ 0.139	0.964 $\pm$ 0.013
	Medium	0.753 $\pm$ 0.148	0.780 $\pm$ 0.010	-1.667 $\pm$ 0.225	0.000 $\pm$ 0.000	0.003 $\pm$ 0.012	0.006 $\pm$ 0.021	0.078 $\pm$ 0.127	0.965 $\pm$ 0.012
	Large	0.753 $\pm$ 0.148	0.780 $\pm$ 0.010	-1.662 $\pm$ 0.222	0.000 $\pm$ 0.000	0.003 $\pm$ 0.012	0.006 $\pm$ 0.021	0.080 $\pm$ 0.141	0.965 $\pm$ 0.012
	Overall	0.748 $\pm$ 0.148	0.780 $\pm$ 0.010	-1.665 $\pm$ 0.221	0.000 $\pm$ 0.000	0.003 $\pm$ 0.012	0.006 $\pm$ 0.021	0.081 $\pm$ 0.135	0.965 $\pm$ 0.013
Gemma	Small	0.728 $\pm$ 0.151	0.786 $\pm$ 0.009	-1.751 $\pm$ 0.216	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.007 $\pm$ 0.023	0.085 $\pm$ 0.139	0.965 $\pm$ 0.012
	Medium	0.737 $\pm$ 0.146	0.801 $\pm$ 0.026	-1.752 $\pm$ 0.225	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.007 $\pm$ 0.023	0.079 $\pm$ 0.127	0.966 $\pm$ 0.011
	Large	0.736 $\pm$ 0.146	0.832 $\pm$ 0.014	-1.745 $\pm$ 0.222	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.008 $\pm$ 0.024	0.081 $\pm$ 0.141	0.967 $\pm$ 0.011
	Overall	0.733 $\pm$ 0.134	0.792 $\pm$ 0.020	-1.750 $\pm$ 0.200	0.000 $\pm$ 0.000	0.004 $\pm$ 0.014	0.007 $\pm$ 0.023	0.082 $\pm$ 0.135	0.966 $\pm$ 0.011

Table 13: Average Sim(LLMGen, Caption) ($\pm$ std) for SBERT, BERTScore F1, BLEURT, BLEU, ROUGE-L, METEOR, and Simpson diversity deltas by caption length. $\Delta\text{Simpson}_{LC}$ shows lexical diversity difference between LLM output and caption; $\Delta\text{Simpson}_{LO}$ is with original reference.



Llama

Mistral

Gemma

Figure 5: Pearson correlation heatmaps between SBERT score and token statistics for different models using BAAI/bge-large-en-v1.5 embeddings.

Table content	table caption	referencing_paragraphs_cleaned	Generated text by LLM	Score																																													
<table><tr><th>Approach</th><th>% input properties covered</th></tr><tr><td>DocTer</td><td>15%</td></tr><tr><td>DeepRel</td><td>16%</td></tr><tr><td>SKIPFUZZ</td><td>31%</td></tr></table>	Approach	% input properties covered	DocTer	15%	DeepRel	16%	SKIPFUZZ	31%	Coverage of input properties	Input diversity. Table shows the experimental results. The inputs used by in its test cases cover two times more input properties than the inputs used in the test cases generated by and. While the inputs selected by and cover only 16% and 15% of the possible input properties, achieves an average of 37% property coverage. This suggests that the diversity of the inputs contributed to the stronger performance of.	This table compares the coverage of input properties among three different approaches: DocTer, DeepRel, and an unnamed tool. It shows the percentage of input properties covered by each approach, with the unnamed tool covering the highest percentage at 31%.	4																																					
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<table><tr><th></th><th>Yes</th><th>No</th><th>Other</th><th>All</th></tr><tr><td>Prior (Goyal et al., 2017)</td><td>61.20</td><td>0.36</td><td>1.17</td><td>25.98</td></tr><tr><td>Language-only (Goyal et al., 2017)</td><td>67.01</td><td>31.55</td><td>27.37</td><td>44.26</td></tr><tr><td>MCH (Paluri et al., 2016)</td><td>78.42</td><td>38.29</td><td>53.86</td><td>62.27</td></tr><tr><td>RUFD (Anderson et al., 2017)</td><td>82.20</td><td>43.90</td><td>56.26</td><td>65.32</td></tr><tr><td>VQA-E (Anderson et al., 2017)</td><td>83.22</td><td>43.58</td><td>56.79</td><td>66.31</td></tr><tr><td>Beyond Bilinear (Ye et al., 2018)</td><td>84.50</td><td>45.39</td><td>59.01</td><td>68.09</td></tr><tr><td>our single</td><td>84.69</td><td>46.75</td><td>59.30</td><td>68.37</td></tr><tr><td>our ensemble-10</td><td>86.15</td><td>47.41</td><td>60.41</td><td>69.66</td></tr></table>		Yes	No	Other	All	Prior (Goyal et al., 2017)	61.20	0.36	1.17	25.98	Language-only (Goyal et al., 2017)	67.01	31.55	27.37	44.26	MCH (Paluri et al., 2016)	78.42	38.29	53.86	62.27	RUFD (Anderson et al., 2017)	82.20	43.90	56.26	65.32	VQA-E (Anderson et al., 2017)	83.22	43.58	56.79	66.31	Beyond Bilinear (Ye et al., 2018)	84.50	45.39	59.01	68.09	our single	84.69	46.75	59.30	68.37	our ensemble-10	86.15	47.41	60.41	69.66	Comparison of our results in VQA task with the state-of-the-art VQA methods on validation set and test-standard set. Accuracies in percentage are reported.	As demonstrated in Table, our system outperforms the state-of-the-art systems which indicates the effectiveness of including caption features as additional inputs. In particular, we observe that our single model outperform other methods especially in 'Num' and 'Other' categories. The reasons are that the additional captions are capable of providing more numerical clues for inferring 'Num' type questions and more common knowledge for answering the 'Other' type questions. Furthermore, an ensemble of 10 single models results in 69.7% in the test-standard split.	This table compares the performance of different methods on a visual question answering (VQA) task. It shows the accuracy of each method in percentage, with our method (both single and ensemble versions) achieving the highest accuracy across all categories. The categories include yes/no questions, numerical questions, and other types of questions, as well as an overall score.	4
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	near	midrange	far	large	medium	small																																											
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	focal	threshold	mixing	mIoU	mIoU ⁺																																												
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<table><tr><th>Method</th><th>HMDB-51</th></tr><tr><td>Network-1 with 4000 background synthetic videos</td><td>72.3 %</td></tr><tr><td>Network-2 with 4000 simplified synthetic videos</td><td>71.8 %</td></tr><tr><td>Network-2 with 8000 simplified synthetic videos</td><td>72.4 %</td></tr></table>	Method	HMDB-51	Network-1 with 4000 background synthetic videos	72.3 %	Network-2 with 4000 simplified synthetic videos	71.8 %	Network-2 with 8000 simplified synthetic videos	72.4 %	Network 2 versus Network 1: note that domain randomization is effective in simulating real outdoor data	In this experiment we tested the effect of using quality versus quantity videos. For Network-1, we added 4000 synthetic videos with an appropriate background setup. For Network-2, we added 4000 simplified videos and in another instance, we added 8000 simplified videos. What we observed (Table) was that adding 8000 videos to Network-2 produced slightly better results than Network-1. Even though Network-1 includes an additional stream compared to Network-2, the latter performed better. We concluded from this experiment that a large number of simplified videos can outperform videos with background included.	This table compares the performance of two different networks, Network 1 and Network 2, on a task involving HMDB-51 data. The results show how the networks perform when trained on different amounts of synthetic video data, with Network 2 showing slightly better performance when trained on more data. The table suggests that using simplified synthetic videos can be effective in simulating real outdoor data.	4																																					
Method	HMDB-51																																																
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Figure 6: Sample outputs generated by LLMs for the table-to-text generation task along with human evaluation score.